

Computación y Estructuras Discretas III

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1 Presentación

2 Regular languages and Automata theory

- Alphabets, strings and languages
- Exercises

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What is an alphabet?

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Which are some examples of alphabets?

- The binary alphabet: $\Sigma = \{0, 1\}$
- The set of all lower case vowels: $\Sigma = \{a, e, i, o, u\}$.
- The set of all ASCII characters, or the set of all printable ASCII characters.

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Which are some examples of strings?

- 01101 is a string from the binary alphabet $\Sigma = \{0, 1\}$
- *banana* is a string from the all lower case letters' alphabet $\Sigma = \{a, b, c, \dots, z\}$.

Example

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aba

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Here we have strings from Σ :

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We can see that $aba \neq aab$, due to the fact that the order of symbols is important when dealing with strings, since strings work as sequences.

What is the empty string?

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Definition

The empty string is the string with zero occurrences of symbols. This string is denoted by λ or ϵ . This string may be chosen from any alphabet whatsoever.

What about the length of a string?

What about the length of a string?

- It is useful to classify strings by their length.
- It is common to say that the length of a string is the number of symbols in the string.
- Not strictly correct. 10101 has only two symbols but there are five positions for symbols (its length is 5).
- Generally expect that the number of symbols can be used when. number of positions is meant.
- The standard notation for the length of a string w is $|w|$.

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$|aba| = 3$, $|baaa| = 4$, $|\lambda| = 0$.

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- We can express the set of all strings of a certain length from that alphabet by using an exponential notation.
- We define Σ^k to be the set of strings of length k , each of whose symbols is in Σ .

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Example

- $\Sigma^0 = \{\lambda\}$ *regardless of what alphabet Σ is.*
- *If $\Sigma = \{0, 1\}$, then $\Sigma^1 = \{0, 1\}$, $\Sigma^2 = \{00, 01, 10, 11\}$.*

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- 1 $u \cdot \lambda = \lambda \cdot u = u$
- 2 $u \cdot (va) = (u \cdot v)a$

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$$u^R = \begin{cases} \lambda & \text{if } u = \lambda \\ a_n \cdots a_2 a_1 & \text{if } u = a_1 a_2 \cdots a_n. \end{cases}$$

What is a substring?

Alphabets, strings and languages

What is a substring?

Definition

A string v is a substring or subword of u if there exist strings x, y such that $u = xvy$. Let's point out that x or y might be λ and therefore, the empty string is a substring of any string and any string is a substring of itself.

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Example

Given $\Sigma = \{a, d, c, d\}$ and $u = bcbaadb$.

<i>Prefixes of u :</i>	<i>Suffixes of u :</i>
λ	λ
b	b
bc	db
bcb	adb
$bcba$	$aadb$
$bcbaa$	$baadb$
$bcbaaad$	$cbaadb$
$bcbaadb$	$bcbaadb$

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$L = \emptyset$, *the empty language.*

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Any language L fulfills $\emptyset \subseteq L \subseteq \Sigma^*$, and it can be either finite or infinite.
We represent languages with upper case letters $A, B, C, \dots, L, M, N, \dots$

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- $\Sigma = \{a, b, c\}$. $L = \{a, aa, aaa, \dots\} = \{a^n \mid n \geq 1\}$.

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- $\Sigma = \{a, b, c\}$. $L = \{a, aba, aca\}$.
- $\Sigma = \{a, b, c\}$. $L = \{a, aa, aaa, \dots\} = \{a^n \mid n \geq 1\}$.
- $\Sigma = \{a, b, c\}$. $L = \{\lambda, aa, aba, ab^2a, ab^3a, \dots\} = \{ab^n a \mid n \geq 0\} \cup \{\lambda\}$.

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Definition

Since languages over Σ are subsets of Σ^ , the usual operations between sets are also valid operations between languages. Then, if A and B are languages over Σ ($A, B \subseteq \Sigma^*$), then the following are also languages over Σ :*

$A \cup B$ Union

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These operations between languages are called boolean operations in order to differentiate them from the linguistic operations, which are extensions to the languages of the operations between strings.

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In general, $AB \neq BA$

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- 3 *Associative property,*

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C.$$

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$$\begin{aligned} A \cdot (B \cup C) &= A \cdot B \cup A \cdot C. \\ (B \cup C) \cdot A &= B \cdot A \cup C \cdot A. \end{aligned}$$

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- 5 *Generalized distributivity property. If $\{B_i\}_{i \in I}$ is any family of languages over Σ , then*

$$\begin{aligned} A \cdot \bigcup_{i \in I} B_i &= \bigcup_{i \in I} (A \cdot B_i), \\ \left(\bigcup_{i \in I} B_i \right) \cdot A &= \bigcup_{i \in I} (B_i \cdot A). \end{aligned}$$

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- Exercises

Exercise

*Given $\Sigma = \{a, b, c, d\}$ define Σ^**

Exercise

Given $u \in \Sigma^$ and $n \in \mathbb{N}$, give a recursive definition of u^n .*

Exercise

Given $u \in \Sigma^$ and $a \in \Sigma$ give a recursive definition of $|u|$.*

Exercise

Given $u \in \Sigma^$ and $a \in \Sigma$ give a recursive definition of u^R .*

Exercise

If $u, v \in \Sigma^$, show that $(uv)^R = v^R u^R$.*

Exercise

Show that $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.

Exercise

Show that $A \cdot \{\lambda\} = \{\lambda\} \cdot A = A$.

Exercise

Show that $A \cdot \emptyset = \emptyset \cdot A = \emptyset$.